Chapter 11 — Introduction to self-supervised learning

Weak supervision still invents labels

Learning objectives

- By the end of this part

Definition versus supervised learning

- Supervised learning trains on input–label pairs, such as images tagged cat or dog
- Self-supervised learning does not require manual task labels during pretraining
- Instead, the model creates its own supervisory signal from the data by solving a pretext

Why pretext tasks matter

- Pretext tasks force the model to learn useful features
- After pretraining
- This reduces labeling cost and can improve generalization when unlabeled corpora are

Vision pretext examples

- Rotation prediction of randomly rotated images
- Solving these proxies teaches spatial and semantic structure that later supports detection

Language pretext examples

- In natural language processing
- Next-sentence prediction and text infilling teach discourse and local coherence
- These tasks turn large unlabeled text corpora into representation learners before any

Contrastive learning idea

- Contrastive methods learn representations by pulling similar views of the same instance
- In vision, two augmented crops of one image form a positive pair
- The resulting encoder captures discriminative features that transfer well when a linear

Example 11.15 — Contrastive Pretraining Then Fine-Tune

- Example 11.15 — hands-on module
- Example 11.15 illustrates contrastive pretraining before fine-tuning
- A vision encoder is pretrained with SimCLR-style contrastive loss on unlabeled product
- Explore the chapter example module
- View files: `modules/chapter11/example15/`

Takeaways

- Self-supervised learning creates supervision from unlabeled structure via pretext tasks
- It sits between supervised and unsupervised learning
- Contrastive pretraining plus light fine-tuning is a practical recipe when downstream

Next

- Complete the quiz for this part
- Time-series forecasting, emphasizing data efficiency when downstream labels remain scarce